

Software Requirements Specification and Architectural Blueprint: AI & LMS-Enabled Cooperative Capacity Building, ERP, and Employment Ecosystem

Executive Summary and Institutional Governance Context

The National Council for Cooperative Training (NCCT), functioning as an autonomous society under the Ministry of Cooperation, Government of India, serves as the apex body for human resource development, executive education, and grassroots capacity building across the Indian cooperative sector¹. Established under the Societies Registration Act of 1860, the council oversees a national training infrastructure dedicated to professionalizing the cooperative movement¹. This network is anchored by the Vaikunth Mehta National Institute of Cooperative Management (VAMNICOM) at Pune, which serves as the apex center for senior leadership education and agribusiness management diplomas¹. Supporting this apex institution are five Regional Institutes of Cooperative Management (RICMs) located at Gandhinagar, Chandigarh, Bengaluru, Kalyani, and Patna, alongside fourteen state-level Institutes of Cooperative Management (ICMs) situated in Bhopal, Bhubaneswar, Chennai, Dehradun, Guwahati, Hyderabad, Imphal, Jaipur, Kannur, Lucknow, Madurai, Nagpur, Pune, and Thiruvananthapuram¹. In addition to these primary institutes, the council provides curriculum oversight and academic guidance to approximately 109 Junior Cooperative Training Centres (JCTCs) operated by state cooperative unions⁴.

Collectively, this nationwide network conducts over 3,200 training programs annually, reaching more than 200,000 participants spanning Primary Agricultural Credit Society (PACS) secretaries, dairy cooperative committee members, Self-Help Groups (SHGs), urban cooperative bankers, women entrepreneurs, and rural youth¹. However, the institutional environment is undergoing a historic structural transformation driven by the Ministry of Cooperation's national agenda⁵. A foundational priority of this agenda is the comprehensive computerization of over 100,000 PACS onto a shared Enterprise Resource Planning (ERP) platform utilizing the standardized Common Accounting System (CAS) and Management Information Systems (MIS)⁶. Concurrently, the establishment of more than 10,000 new multi-purpose PACS, dairy societies, and fisheries federations across panchayats has created an immediate demand for continuous, verified, and standardized operational training⁶. Historically, training operations across VAMNICOM, the RICMs, and ICMs have relied on

decentralized, manual, and disconnected administrative workflows¹. Course scheduling, nomination processing, and trainee registrations are frequently managed through disparate spreadsheets, paper ledgers, or siloed portal instances¹⁰. Attendance verification in remote training units depends almost entirely on physical signature sheets, introducing risks of proxy attendance and inaccurate trainee reporting¹⁰. E-learning repositories remain static, web-centric, and predominantly mono- or bilingual, preventing effective knowledge transfer to rural trainees who require localized instruction in their native languages¹. Furthermore, course certifications are printed on conventional paper substrates without cryptographic protection, making instant verification difficult for hiring organizations¹⁰. Crucially, an institutional gap separates training completion from rural employment: while cooperative banks, federations, and agro-processing hubs actively seek qualified personnel, certified rural youth lack a unified platform to showcase verified competencies¹⁰.

The proposed digital ecosystem resolves these operational challenges by deploying an enterprise cloud-based platform that unifies academic ERP functions, an offline-capable multilingual Learning Management System (LMS), edge biometric attendance capture, cryptographic credential verification via DigiLocker, and an AI-driven career guidance and employment exchange¹⁰.

Operational Dimension	Legacy Institutional Architecture	Target Unified Ecosystem
Institutional Governance	Fragmented spreadsheets and decentralized scheduling across 20 distinct institutes ¹ .	Centralized multi-tenant academic ERP with automated capacity optimization and dynamic calendar generation ¹⁰ .
Participant Identity Management	Isolated, batch-specific paper rosters with substantial data duplication ¹⁰ .	Unified Trainee Profile indexed to unique civil identifiers with lifelong competency tracking ¹⁰ .
Attendance Verification	Manual sign-in sheets susceptible to proxy attendance and reporting delays ¹⁰ .	Edge AI facial verification and dynamic rotating QR codes with cryptographically signed local buffers ¹⁰ .
Pedagogical Delivery Model	Physical classroom sessions and static desktop repositories ¹ .	Offline-first Progressive Web Application (PWA) and mobile client with

		automated delta sync protocols ¹⁰ .
Linguistic Localization	Instructional materials limited primarily to English and Hindi ¹ .	Automated localization across 22 Scheduled Languages via MeitY's Bhashini ULCA pipeline ¹⁵ .
Credential Lifecycle	Unverifiable physical certificates requiring manual administrative validation ¹⁰ .	W3C Verifiable Credentials issued directly to citizen DigiLocker wallets via API Setu ¹⁰ .
Employment Linkages	Disconnected recruitment relying on informal local notices without skill verification ¹⁰ .	Semantic vector-based job-matching engine integrated with the National Career Service (NCS) ¹⁰ .

System Conceptual Framework and Operational Boundaries

The system conceptualizes the capacity-building ecosystem as an integrated lifecycle that tracks the rural learner from initial course discovery through to productive placement in the cooperative economy¹⁰. The platform coordinates activities across four operational layers:

1. Central Policy and Planning Layer: Allows the Ministry of Cooperation and NCCT headquarters to establish strategic training targets, track state-level implementation, monitor program budgets, and review national analytics².
2. Institutional ERP and Campus Logistics Layer: Automates candidate nomination screening, batch formation, faculty allocation, and hostel accommodation management across VAMNICOM, RICMs, and ICMs¹.
3. Pedagogical and Identity Delivery Layer: Delivers interactive, multilingual micro-learning modules to edge mobile terminals while tracking academic progress through offline-tolerant mechanisms and verifying attendance via edge biometrics¹⁰.
4. Extension, Verification, and Employment Exchange Layer: Converts verified academic achievements into tamper-evident credentials, providing rural candidates with contextual career guidance and directly linking certified talent with cooperative societies¹⁰.

The operational boundary of the platform includes all functional requirements associated with program administration, participant lifecycle management, digital learning delivery, edge identity verification, credential issuance, and job discovery¹⁰. Conversely, the platform maintains clear interface boundaries: it does not function as a core banking system for cooperative banks, nor does it replace the transaction accounting engines embedded within computerized PACS⁶. External systems—including the National Cooperative Database (NCD),

the Skill India Digital Hub (SIDH), DigiLocker, and the National Career Service (NCS)—interoperate exclusively through authenticated API gateways, preserving architectural modularity⁶.

Persona Category	Institutional Role	Operational Permissions and System Boundaries
Trainee / Rural Youth	Candidate, Student, Jobseeker ¹⁰ .	Discovers training programs; submits enrollment requests; completes offline mobile courses; marks biometric attendance; accesses DigiLocker credentials; uses the AI career chatbot; applies for cooperative vacancies ¹⁰ .
Training Administrator	Program Coordinator at RICM/ICM ¹ .	Configures batch calendars; validates organizational nominations; assigns classrooms and faculty; oversees residential hostel allotments; reviews daily attendance logs; approves certificate generation ¹⁰ .
Faculty / Resource Person	Subject Matter Expert / Trainer ¹⁰ .	Authors multilingual curriculum modules; creates interactive assessments; monitors learner progression; generates dynamic session QR codes; reviews attendance overrides ¹⁰ .
Institutional Director	Head of VAMNICOM, RICM, or ICM ¹ .	Evaluates campus-wide operational metrics; tracks resource and bed utilization; monitors training targets set by the Ministry

		of Cooperation; audits institutional finances and honoraria ¹ .
Cooperative Employer	HR / Board Member at PACS, DCCB, Dairy Union ⁶ .	Publishes job requisitions; defines prerequisite competency profiles; filters verified certified candidates; accesses authenticated digital credentials; manages interview pipelines ¹⁰ .
Central NCCT / Ministry Admin	Executive at Ministry of Cooperation / NCCT HQ ¹ .	Oversees national multi-tenant configurations; monitors aggregate Key Performance Indicators (KPIs); evaluates cross-state performance; manages master data and regulatory compliance ² .

Architectural Topology and Multi-Tier Framework

The platform employs a cloud-native, distributed microservices architecture designed to maintain performance, fault isolation, and horizontal scalability under fluctuating regional network conditions¹⁰. The system handles central administrative loads through containerized microservices in cloud environments, while offloading high-frequency identity and content delivery operations to local edge instances deployed at training centers¹⁰.

The presentation tier uses a responsive web application built with modern component frameworks for administrative, faculty, and employer dashboards¹⁰. Trainees and field supervisors interface through a cross-platform mobile application built on Flutter, utilizing an embedded SQLite database (managed via the Room abstraction layer) for offline transaction buffering¹⁰. Classroom attendance stations and hostel kiosks operate on low-power ARM-based devices running embedded Linux or Android runtimes, executing local facial recognition pipelines directly on-device¹⁰.

Incoming traffic routes through an enterprise API Gateway that manages TLS 1.3 termination, rate limiting, and request routing¹⁰. Authentication uses an OpenID Connect (OIDC) identity provider, issuing cryptographically signed JSON Web Tokens (JWT) that encode tenant identifiers, user roles, and fine-grained permissions¹⁰. Communication across core services runs over containerized application meshes, using asynchronous message brokers to decouple long-running operations—such as multi-lingual media transcoding, verifiable credential

compilation, and analytics processing—from user-facing request threads¹⁰. The intelligence and language tier integrates with MeitY's Bhashini Universal Language Contribution APIs (ULCA) to orchestrate speech recognition, neural machine translation, and speech synthesis workflows across 22 Scheduled Languages¹⁵. Contextual career guidance executes via a Retrieval-Augmented Generation (RAG) pipeline: user queries are converted into dense vector embeddings and matched against curated curriculum guides and cooperative policy documentation, ensuring generated guidance remains accurate and grounded⁶. The persistence tier utilizes PostgreSQL with the pgvector extension for transactional data and vector storage, an in-memory Redis cluster for session states and ephemeral one-time-password caches, and S3-compatible distributed object storage for binary media assets¹⁰.

Architectural Tier	Selected Technology Components	Operational Role and Domain Boundary	Fault-Tolerance and High-Availability Strategy
Client & Edge Tier	Flutter, SQLite, Next.js, Embedded Linux/Android Runtimes ¹⁰ .	User interface rendering, client validation, local media storage, on-device facial feature extraction ¹⁰ .	Offline-first local operation; client transaction buffering during wide-area network (WAN) outages ¹⁰ .
Ingress & Security Tier	Enterprise API Gateway, OIDC Provider, WAF, Redis Rate Limiter ¹⁰ .	Transport layer security, token introspection, DDoS protection, traffic shaping, API route routing ¹⁰ .	Multi-Availability-Zone deployment with automated failover and dynamic health probing.
Core Microservices Mesh	FastAPI, Go, Docker, Kubernetes Orchestration ¹⁰ .	Executes core business logic for ERP, LMS, logistics, credential issuance, and recruitment modules ¹⁰ .	Horizontal Pod Autoscaling (HPA) governed by CPU saturation and request queue latency ¹⁰ .
Intelligence & Speech Tier	Bhashini ULCA APIs, LangChain, pgvector, PyTorch	Indic voice transcription, neural machine	Asynchronous message queues with Redis backoff

	Runtime ¹⁵ .	translation, RAG career counseling, semantic candidate matching ¹⁰ .	retry loops and local translation caching ¹⁰ .
Persistence & Storage Tier	PostgreSQL 16+, Redis Sentinel Cluster, S3 Object Storage ¹⁰ .	ACID relational persistence, vector embedding indexing, ephemeral TOTP caches, unstructured binary storage ¹⁰ .	Multi-region synchronous database replication, automated WAL archiving, and periodic snapshot restoration tests ¹⁰ .

Detailed Functional Requirements

Programme Administration, Society Nomination, and Scheduling Engine

The academic ERP subsystem digitizes administrative and logistical workflows across all twenty NCCT institutions and affiliated junior centers¹. The system provides role-based workflows for course creation, enrollment verification, and batch lifecycle tracking¹⁰.

The operational lifecycle begins with the publication of the Annual Training Calendar¹. Central administrators at NCCT headquarters and regional directors at VAMNICOM, RICMs, and ICMs configure approved programs, defining curriculum frameworks, instructional languages, intake capacity, and target stakeholder groups (such as PACS secretaries, Board of Directors members, or rural youth)¹. The platform supports individual self-enrollment as well as bulk organizational nominations from sponsoring cooperative bodies, including District Central Cooperative Banks (DCCBs), State Cooperative Agriculture and Rural Development Banks (SCARDBs), and dairy federations¹. When processing organizational nominations, the engine verifies candidate eligibility against historical records, ensuring prerequisite qualifications are met while flagging duplicate enrollments¹⁰.

Once enrollment closes, the batch management engine organizes participants into training cohorts. A resource scheduling algorithm maps these cohorts to lecture halls, computer labs, and trainers, resolving scheduling bottlenecks and resource conflicts across overlapping courses¹⁰. Any scheduling updates automatically update the calendar views of assigned faculty and registered trainees, accompanied by automated notifications delivered through mobile push channels, SMS gateways, and in-app alerts¹⁰.

Campus Logistics, Facilities, and Hostel Allocation Subsystem

Residential training programs at VAMNICOM, the five RICMs, and fourteen ICMs require coordinated management of physical campus resources¹. The logistics subsystem automates room reservations, hostel allocations, and dining operations¹⁰.

The hostel management engine maintains an inventory of residential assets, categorizing rooms by capacity, building wing, maintenance status, and accessibility accommodations. To optimize bed allocation upon batch finalization, the system employs an assignment model designed to maximize room occupancy while respecting gender segregation rules, scheduling boundaries, and institutional quotas:

$$\min \sum_{i \in \mathcal{T}} \sum_{j \in \mathcal{R}} c_{ij} X_{ij}$$

subject to:

$$\sum_{j \in \mathcal{R}} X_{ij} = 1 \quad \forall i \in \mathcal{T}, \quad \sum_{i \in \mathcal{T}} X_{ij} \leq \text{Capacity}_j \quad \forall j \in \mathcal{R}, \quad X_{ij} \in \{0, 1\}$$

In this formulation, $\mathcal{T}$ represents the set of enrolled trainees, $\mathcal{R}$ denotes the set of available residential rooms, Capacity_j reflects the bed limit of room j , and c_{ij} represents penalty costs associated with sub-optimal assignments (such as placing trainees from different courses or schedules into shared quarters).

The platform also tracks dining mess operations, monitoring daily meal headcounts, tracking special dietary requirements, and managing food service billing. These logs integrate with institutional finance workflows to reconcile per-diem allowances, track daily trainee stipends, and automate sponsorship invoicing for participating cooperative societies¹.

Edge Biometric Attendance and Cryptographic Verification

To prevent attendance fraud and verify participant presence across both campus lecture halls and decentralized field workshops, the platform deploys a dual-mode verification framework combining edge facial recognition and dynamic QR codes¹⁰.

The facial verification pipeline is engineered for on-device execution on edge tablets and mobile devices, avoiding dependencies on continuous internet connectivity¹⁰. During program onboarding, the platform captures the trainee's facial features, executes an edge-optimized deep convolutional neural network (MobileFaceNet with ArcFace margin loss), and generates a 512-dimensional normalized embedding vector:

$$\mathbf{e} \in \mathbb{R}^{512}, \quad \|\mathbf{e}\|_2 = 1$$

Raw facial images are deleted immediately after vector extraction¹³.

During subsequent verification events, the edge terminal captures a live video feed, confirms physical presence via passive texture evaluation and active challenge prompts (such as

randomized blink or head-turn checks), extracts the live embedding vector $\mathbf{e}_{\text{live}}$, and

computes the cosine similarity against the locally cached template $\mathbf{e}_{\text{template}}$ ¹⁰:

$$S_{\cos}(\mathbf{e}_{\text{live}}, \mathbf{e}_{\text{template}}) = \mathbf{e}_{\text{live}} \cdot \mathbf{e}_{\text{template}}$$

A match is confirmed when $S_{\cos} \geq 0.68$, meeting biometric matching requirements while maintaining low False Acceptance Rates¹⁰.

In parallel, the platform provides a dynamic QR code alternative for classrooms lacking biometric hardware¹⁰. The instructor's display terminal generates an ephemeral QR code that refreshes every 15 seconds, encoding a Time-Based One-Time Password (TOTP) signed via HMAC-SHA256:

$$\text{Payload} = \text{BatchID} \parallel \text{SessionID} \parallel T_k \parallel \text{HMAC}_K(\text{BatchID} \parallel \text{SessionID} \parallel T_k)$$

Here, $T_k = \lfloor t_{\text{epoch}} / 15 \rfloor$ represents the time-step counter and K is a cryptographic key shared between the institutional server and the terminal. The trainee scans the code using the native mobile application, which validates that the trainee's device clock and GPS/NavIC coordinates fall within the geofenced training venue before registering the attendance mark¹⁰. If wide-area network access is unavailable, attendance records are signed with the terminal's private key, stored in a local SQLite encrypted buffer, and automatically synchronized over an asynchronous queue when network access returns¹⁰.

Offline-First Multilingual LMS and Assessment Engine

The Learning Management System provides continuous instructional delivery across rural regions characterized by variable bandwidth and diverse languages¹⁰. The architecture employs an offline-first execution model: trainees can download SCORM 2004 modules, micro-lecture audio files, and interactive HTML5 exercise packages over local campus Wi-Fi or public broadband points for offline review¹⁰.

Trainee progress—including lesson completions, media playback positions, assessment attempts, and time-on-task metrics—is captured locally as structured xAPI (Experience API) statements within the device's encrypted SQLite database¹⁰. When the mobile client reconnects to the network, a delta synchronization protocol reconciles local logs with the central server. The system uses Lamport Timestamps and Conflict-Free Replicated Data Types (CRDTs) to ensure state updates merge monotonically:

$$\text{State}_{\text{server}} \leftarrow \max(\text{State}_{\text{server}}, \text{State}_{\text{client}})$$

This mathematical approach prevents progress rollbacks and prevents newer offline progress logs from being overwritten by older server snapshots¹⁰.

Linguistic accessibility is handled through an architectural integration with the Bhashini Universal Language Contribution APIs (ULCA)¹⁵. Rather than relying entirely on manual course translation, training materials are routed through Bhashini's Neural Machine Translation (NMT) pipelines to generate text in 22 Scheduled Languages¹⁵. To accommodate learners with limited functional literacy, the LMS incorporates Bhashini's Automated Speech Recognition (ASR) and

Text-to-Speech (TTS) engines, allowing users to listen to curriculum materials and submit oral responses in their native dialects¹⁵. The assessment module delivers both formative quizzes and final certification examinations, supporting randomized question order, anti-cheating time limits, and automated remedial recommendations for topics where trainees demonstrate low comprehension¹⁰.

Verifiable Skill Credentials and National Ecosystem Ingestion

To address credential fraud and eliminate manual certificate verification across the cooperative sector, the platform issues cryptographically signed digital credentials compliant with the W3C Verifiable Credentials Data Model 2.0¹⁰.

When a trainee satisfies both program completion criteria (requiring at least 80% attendance) and assessment standards (achieving a score of at least 75%), the system compiles a standardized JSON-LD credential document¹⁰. This document details the trainee's verified identifier, course competency outcomes, National Occupational Standards (NOS) codes, and institutional accreditation details¹. The document is digitally signed using an Ed25519 asymmetric cryptographic keypair held within NCCT's cloud Hardware Security Module (HSM), appending a tamper-evident proof (Ed25519Signature2020)¹⁸.

The credential engine implements the national DigiLocker Issuer API specification¹¹. Upon certificate signing, metadata containing the trainee's identifier and certificate reference is pushed via API Setu to the central repository¹². When a trainee views their DigiLocker mobile wallet, the application queries the NCCT Pull URI endpoint to fetch, verify, and render the signed credential¹¹.

For physical verification, printed certificates display a high-density, cryptographically signed 2D QR code¹⁰. Employers and inspecting officers can scan this QR code using standard smartphone cameras to resolve the credential against the NCCT Decentralized Identifier (DID) registry, verifying authenticity without exposing unnecessary personal information¹⁰.

Conversational Career Advisory and Semantic Employment Exchange

To bridge the gap between technical skill certification and rural employment, the platform integrates an AI-driven career guidance chatbot and a recruitment matching engine¹⁰.

The career guidance chatbot is built as an Indic conversational agent that accepts both text and voice queries through Bhashini speech-to-text integration¹⁰. To prevent the generation of inaccurate or speculative career advice, the model uses Retrieval-Augmented Generation (RAG)¹⁰. The conversational pipeline indexes NCCT curriculum catalogs, the PACS HR Policy framework, Model Bye-Laws for Multi-Purpose PACS, and operational manuals for dairy and credit cooperatives into a vector database¹. When a rural learner asks about career opportunities, the engine retrieves matching policy documents, passes them as context to a language model, and delivers practical, actionable guidance—such as explaining the qualifications required to become a PACS secretary, cooperative accountant, or dairy quality manager⁶.

The recruitment subsystem provides a verified candidate portal for cooperative employers, including PACS, District Central Cooperative Banks, State Cooperative Banks, and

agro-processing federations⁵. Recruiters can define vacancy profiles by specifying required skill sets, certification prerequisites, and geographic location preferences¹⁰. The platform computes a candidate match score using a multi-factor matching function:

$$S(c, j) = w_{\text{skill}} S_{\text{semantic}}(\mathbf{v}_c, \mathbf{v}_j) + w_{\text{cert}} C(c, j) + w_{\text{exp}} E(c, j) + w_{\text{geo}} G(c, j)$$

where weights are normalized such that:

$$w_{\text{skill}} + w_{\text{cert}} + w_{\text{exp}} + w_{\text{geo}} = 1.0$$

The component functions evaluate candidate qualifications systematically:

- $S_{\text{semantic}}(\mathbf{v}_c, \mathbf{v}_j) \in [0, 1]$ represents the cosine similarity between the candidate's verified skill vector $\mathbf{v}_c$ and the employer's job description vector $\mathbf{v}_j$, computed using domain-adapted sentence embeddings.
- $C(c, j) \in [0, 1]$ represents the proportion of required NCCT skill certifications verified against the candidate's profile¹⁰.
- $E(c, j) \in [0, 1]$ normalizes relevant agricultural and cooperative sector experience against employer requirements.
- $G(c, j) = \exp(-\lambda \cdot d(c, j))$ applies an exponential distance decay over the geodesic distance $d(c, j)$ between the candidate's home district and the workplace, prioritizing local rural youth for community cooperative roles while permitting geographic mobility for specialized leadership posts.

Data Schemas and Technical Specifications

The persistence architecture utilizes PostgreSQL 16 with the pgvector extension to handle structured relational entities and high-dimensional semantic embeddings within an integrated data model¹⁰. The schema enforces data integrity through foreign-key relationships, unique constraints, and spatial indexes.

Entity Table Name	Primary Identifier	Core Foreign Keys and Attributes	Indexing and Relational Constraints
trainee_profiles	trainee_id (UUID)	user_id (UUID), aadhaar_sha256 (VARCHAR(64)), full_name, dob (DATE), gender,	Unique constraint on aadhaar_sha256; composite B-Tree index on (state_id,

		district_id, state_id, education_level, created_at ¹⁰ .	district_id) ¹⁰ .
biometric_templates	template_id (UUID)	trainee_id (UUID), embedding_vector (VECTOR(512)), algorithm_tag (VARCHAR(32)), liveness_score, updated_at.	Foreign Key to trainee_profiles; HNSW index on embedding_vector with cosine distance metric.
training_programmes	programme_id (UUID)	institution_id (UUID), course_code (VARCHAR(32)), title, sector_code, duration_hours (INT), intake_limit (INT), status.	Unique constraint on course_code; B-Tree index on (institution_id, sector_code).
training_batches	batch_id (UUID)	programme_id (UUID), lead_trainer_id (UUID), start_date (DATE), end_date (DATE), facility_id (UUID), batch_state.	Check constraint (end_date >= start_date); B-Tree index on batch_state.
attendance_records	record_id (UUID)	batch_id (UUID), trainee_id (UUID), session_id (UUID), recorded_at (TIMESTAMPTZ), method (ENUM), similarity_score, device_id.	Composite index on (batch_id, session_id, trainee_id); table partitioned by month.
verifiable_credentials	credential_id (UUID)	trainee_id (UUID), batch_id (UUID), certificate_urn	Unique constraint on certificate_urn; B-Tree index on

		(VARCHAR(128)), issued_at, signature_hex, digilocker_uri, status ¹² .	digilocker_uri ¹² .
job_postings	job_id (UUID)	recruiter_id (UUID), society_type, title, skills_array (TEXT[]), district_id, vacancies (INT), requirement_vector (VECTOR(512)).	HNSW vector index on requirement_vector ; GIN index on skills_array.
audit_ledger_events	event_id (UUID)	actor_id (UUID), action_type (VARCHAR(64)), target_entity_id (UUID), client_ip, payload_hash (VARCHAR(64)), timestamp ¹⁰ .	Append-only ledger; immutable table rules; B-Tree index on (timestamp, actor_id) ¹⁰ .

The platform defines standard schemas for data exchange across subsystems, as outlined in the following specification tables.

Specification Field	W3C Verifiable Credential Schema	Bhashini ULCA Pipeline Execution	Edge Attendance Synchronization Batch
Payload Identifier	urn:uuid:8f14b3d7-2194-4e4b-97e3-0d319e7a9c21 ¹⁸ .	Task Request Handshake: pipelineId: NCCT-Indic-Audio-v1 ¹⁶ .	Batch Header: syncBatchId: c4b9281a-6dfb-48ae-94a2-19273c5bb201 ¹⁰ .
Context / Task Definition	https://www.w3.org/2018/credentials/v1 and https://schema.ncct.ac.in/v1 ¹⁸ .	Ordered Task Sequence: [ASR, Translation, TTS] across target language pairs ¹⁶ .	Metadata: edgeDeviceId, batchId, syncTimestamp, terminalSignature ¹⁰ .

Entity / Subject Details	did:india:trainee:992837418234 with Course, Grade, and Competencies ¹⁸ .	Input audio buffer (Base64 WAV, 16kHz) and source dialect configuration ¹⁶ .	Array of verified attendance logs with trainee IDs, timestamps, and confidence scores ¹⁰ .
Security Proof / Checksum	Cryptographic proof: Ed25519Signature2020 signed via NCCT HSM ¹⁸ .	Cryptographic Bearer token issued by National Language Translation Mission ¹⁶ .	Device cryptographic signature verified using ECDSA over SHA-256 payload digest.

National Digital Public Infrastructure (DPI) Interfaces and Integration Framework

The platform integrates directly into India's Digital Public Infrastructure, linking cooperative capacity building with national identity, skilling, and employment registries¹¹.

The DigiLocker and API Setu integration implements the standard digital document issuance framework¹¹. The platform hosts a secure Pull URI endpoint that handles real-time queries initiated by citizens within their DigiLocker wallets¹¹. When a certified trainee links their Aadhaar or APAAR identity inside DigiLocker, the gateway queries the NCCT Pull URI service, which verifies the user token, generates the signed XML/PDF credential, and deposits it into the citizen's repository¹¹. Concurrently, the Push URI endpoint notifies the central directory of newly issued certificates, ensuring immediate document discovery¹².

The Skill India Digital Hub (SIDH) interface provides bidirectional synchronization with the Ministry of Skill Development and Entrepreneurship (MSDE)¹¹. The platform syncs training curriculum frameworks with National Occupational Standards (NOS), links NCVET-accredited course codes, and exports completion records to populate national skilling databases¹¹. This synchronization prevents duplicate training benefits and supports credit transfers within the Academic Bank of Credits (ABC)²⁶.

The National Career Service (NCS) bridge connects cooperative training outputs with the broader labor market¹¹. The platform runs asynchronous integration workers that query the NCS API to ingest rural job vacancies, credit society openings, and agribusiness listings¹¹. In return, the platform publishes verified profiles of certified candidates to the NCS portal, allowing cooperative federations to discover qualified talent across regional districts¹⁰.

The Bhashini Universal Language Contribution APIs (ULCA) integrate language translation throughout the application stack¹⁵. Course curricula, lecture audio, and user queries route through Bhashini endpoints to generate localized translations and synthesized speech across 22 Scheduled Languages, ensuring consistent accessibility for rural users¹⁵.

Interfacing Platform	Communication Protocols	API Architecture & Endpoint Type	Data Flow Direction	Functional Scope & Payload Definition
DigiLocker (API Setu) [cite: 12, 17]	HTTPS REST / Mutual TLS; JSON-LD and XML formats ¹² .	Callback Handshake: Pull URI and Push URI endpoints ¹² .	Bidirectional ¹²	Digital Skill Certifications, Marksheets, Revocation Lists, Verifiable Proofs ¹⁰ .
Skill India Digital Hub (SIDH) [cite: 11, 17]	HTTPS REST; OAuth 2.0 Bearer Tokens; JSON ¹¹ .	Batch Synchronization via secure API Setu channels ¹¹ .	Bidirectional ¹¹	Trainee skilling records, NCVET course codes, trainer accreditation registries ¹¹ .
National Career Service (NCS) [cite: 11]	HTTPS REST; Encrypted Payload; JSON ¹¹ .	Asynchronous Pub/Sub queues and scheduled worker routines ¹⁰ .	Bidirectional ¹¹	Vacancy listings, verified cooperative candidate profiles, interview status tracking ¹⁰ .
Bhashini (NLTM ULCA) [cite: 15, 16]	HTTPS REST / gRPC; API Key Authentication ⁵ .	Synchronous compute pipeline execution (/pipeline/compute) ¹⁶ .	Bidirectional ¹⁶	Audio streams (Base64 WAV), transcribed text strings, translated localized outputs ¹⁵ .
National Cooperative Database (NCD)	HTTPS REST; mTLS Certificate	Scheduled nightly batch synchronization	Inbound	Society registry IDs, PACS geographic

[cite: 5, 6]	Binding; JSON.	n.		locations, registered board profiles ⁵ .
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Hardware Configurations and Edge Deployment Infrastructure

In accordance with the Software + Hardware deployment model, the platform pairs its cloud services with an edge computing architecture¹⁰. This design deploys standard commercial off-the-shelf (COTS) hardware across training centers, providing reliable identity verification and content delivery without requiring proprietary manufacturing¹⁰. Campus Edge Micro-Servers are deployed across VAMNICOM, the five RICMs, fourteen ICMs, and participating junior centers¹. These micro-servers host lightweight container clusters that run local content caches and SQLite sync brokers¹⁰. Video lectures, interactive SCORM files, and audio assets are served directly over local Wi-Fi, minimizing external bandwidth use. During WAN outages, these edge servers maintain continuous classroom attendance tracking and assessment delivery¹⁰.

Classroom Attendance Kiosks utilize ruggedized touch tablets or Single Board Computers (SBCs) paired with dual RGB and Near-Infrared (NIR) imaging sensors¹⁰. The dual-camera setup provides reliable face verification across changing lighting conditions while preventing photo or video playback spoofing attacks¹⁰. The device's integrated Neural Processing Unit (NPU) runs the MobileFaceNet pipeline locally, matching captured faces against pre-loaded batch templates in under 800 milliseconds per person¹⁰.

Mobile Attendance Terminals operate on standard Android smartphones utilized by traveling instructors for remote workshops¹⁰. The application leverages the device's native camera and NavIC/GPS hardware to enforce geofenced facial recognition and dynamic QR code generation, storing encrypted logs locally during offline field sessions¹⁰.

Hostel and Logistics Kiosks are deployed at campus residential halls and dining facilities to verify check-ins, record meal access, and track room occupancy¹⁰.

Hardware Role	Recommended Form Factor	Processing & Memory Specifications	Sensor & Communication Hardware	Institutional Deployment Context
Campus Edge Micro-Server	1U Rackmount or Industrial Small Form Factor (SFF) Unit.	x86-64 8-Core Processor, 32 GB ECC RAM, 1 TB NVMe SSD Storage.	Dual Gigabit Ethernet (RJ-45), Local UPS Battery Link, Hardware TPM 2.0.	On-premise server rooms at VAMNICOM, 5 RICMs, and 14 ICM campus hubs ¹ .

Classroom Biometric Kiosk [cite: 10]	Wall-mounted ruggedized tablet or industrial SBC enclosure ¹⁰ .	ARM Cortex Octa-Core Processor, 4 GB LPDDR4, 64 GB eMMC, $\geq$ TOPS NPU.	5MP RGB Camera, 2MP NIR Camera, IR LED Array, Wi-Fi 802.11ac, Bluetooth 5.0, 4G LTE.	Lecture halls, computer laboratories, and main campus auditoriums ¹⁰ .
Mobile Attendance Terminal [cite: 10]	Commercial Android Smartphone or Tablet (COTS) ¹⁰ .	Quad-Core 2.0 GHz Processor, 3 GB RAM, Android 11+ Operating System ¹⁰ .	8MP Autofocus Camera, NavIC / GPS Positioning Sensor, 4G/5G Cellular Link.	Field instructors conducting training programs at remote PACS sites ¹ .
Logistics & Mess Kiosk [cite: 10]	Desktop POS Terminal or Wall-Mounted Enclosure ¹⁰ .	Quad-Core ARM Processor, 2 GB RAM, 32 GB Storage ¹⁰ .	Wide-Angle QR Camera, Optional Mifare 13.56 MHz RFID Reader, Ethernet / Wi-Fi.	Residential hostel lobbies, dining facilities, and security access points ¹⁰ .

Security Architecture, Cryptography, and Statutory Data Protection Compliance

The platform's security framework implements technical and operational safeguards aligned with the Digital Personal Data Protection (DPDP) Act, 2023, and the DPDP Rules, 2025²⁹. The National Council for Cooperative Training (NCCT) operates as the designated Data Fiduciary, maintaining accountability for personal data processing carried out across its training network²⁹. In compliance with Purpose Limitation principles under Section 6 of the DPDP Act, collected trainee information and biometric records are used strictly for training registration, attendance auditing, and certification, and cannot be processed for external surveillance or commercial marketing²⁹. Prior to capturing any personal information, the application presents a clear, concise notice in the user's chosen language, outlining the specific data gathered, processing purposes, and retention timelines²⁹. User consent is logged as an auditable, timestamped record²⁹. Trainees retain the statutory right to withdraw biometric consent at any point without losing access to training, in which case the system defaults to dynamic QR code attendance verification¹⁰.

The platform enforces strict biometric data minimization to safeguard trainee privacy²⁹. Raw facial images, video recordings, and unencrypted templates are never written to permanent disk storage or transmitted over network links²⁹. During enrollment, live facial images are captured in volatile memory, converted into 512-dimensional floating-point embedding vectors ($\mathbf{e} \in \mathbb{R}^{512}$), and deleted immediately from RAM¹³.

The resulting embedding vectors are salted with tenant-specific keys and encrypted using AES-256-GCM before storage²⁹. Because these mathematical representations cannot be reverse-engineered to reconstruct the original facial image, identity theft risks are minimized. In accordance with Section 8(7) of the DPDP Act, biometric vectors are subject to automated data retention limits: templates are removed from edge verification devices once a training batch concludes and are permanently erased from central databases 90 days after course completion, unless explicit consent is renewed for ongoing certifications³⁰.

To fulfill breach notification obligations under Section 8(6), the platform incorporates automated intrusion detection controls³⁰. Any unauthorized access attempts trigger automated isolation protocols and generate immutable audit logs, while notifying administrative officers and CERT-In within mandated reporting windows¹⁰.

System Quality Attributes and Non-Functional Requirements

The non-functional specifications define performance, reliability, security, and accessibility standards for the system across cloud and edge tiers¹⁰.

Quality Dimension	Quantitative Metric / Standard	Technical Mechanism and Enforcement Strategy
API Response Latency [cite: 10]	$\leq$ for 95th percentile requests; $\leq$ for 99th percentile under peak loads ¹⁰ .	Redis session caching, optimized relational query plans, read-replica connection pools, CDN edge caching ¹⁰ .
Edge Biometric Speed [cite: 10]	$\leq$ per face verification event at local terminals ¹⁰ .	Quantized MobileFaceNet model execution in ONNX Runtime utilizing ARM NEON and local NPU accelerators ¹³ .
Biometric Match Accuracy	$FAR \leq$;	ArcFace margin loss optimization; adaptive

	$FRR \leq$; Liveness Detection $\geq$.	cosine distance thresholding set at $\tau =$ ¹³ .
Platform Availability [cite: 10]	99.9% annual uptime ($\leq$ hours unplanned downtime per year) ¹⁰ .	Multi-Availability-Zone container orchestration, automated Kubernetes failover, hot database standby ¹⁰ .
Offline Durability [cite: 10]	14 days of offline operational continuity with zero local record loss ¹⁰ .	SQLite database with Write-Ahead Logging (WAL), encrypted flash storage, CRDT synchronization protocols ¹⁰ .
System Throughput	Support for 25,000 concurrent learners; 2,500 attendance API transactions per second.	Stateless microservices, horizontal container scaling, asynchronous task processing via message queues ¹⁰ .
Data Encryption [cite: 10]	TLS 1.3 for data in transit; AES-256-GCM for data at rest across all tiers ¹⁰ .	Strict Transport Security (HSTS) enforcement, key management with automated rotation, salted hashing ¹⁰ .
Linguistic Localization [cite: 10]	Complete interface localization across 22 Scheduled Languages; sub-second translation caching ¹⁵ .	Integration with MeitY Bhashini ULCA platform; in-memory caching of localized interface strings ¹⁰ .
Digital Accessibility	WCAG 2.1 Level AA Compliance across all web and mobile interfaces.	High-contrast layouts, full keyboard navigation, screen-reader compatibility, text-to-speech audio alternatives ¹⁵ .

Verification, Testing, and Quality Assurance

Framework

The verification framework establishes systematic validation procedures to confirm functional reliability, biometric accuracy, and offline resilience prior to deployment across NCCT institutions¹⁰.

Automated testing suites validate business logic across core ERP workflows, evaluating enrollment rules, nomination quotas, and scheduling constraints to prevent resource allocation conflicts¹⁰. The biometric attendance subsystem undergoes evaluation against standardized demographic face datasets to ensure balanced accuracy across age, gender, and regional phenotypes¹⁰. Liveness detection algorithms are tested against presentation attack vectors—including printed photographs, mobile video replays, and silicone masks—to confirm resistance against spoofing attempts¹⁰.

Offline resilience is tested through simulated network disruptions: automated proxies inject network latency, packet loss, and full WAN outages while mobile and edge clients capture learning events and attendance records¹⁰. These tests verify that transactions persist in local SQLite storage and merge accurately via CRDT protocols once network connectivity is restored, ensuring zero data loss and preventing duplicate rows¹⁰. Credential generation is validated by checking issued JSON-LD documents against W3C schemas, verifying Ed25519 cryptographic signatures, and testing DigiLocker Pull URI callbacks under simulated loads¹².

QA Evaluation Dimension	Target Component Under Test	Testing Methodology and Tools	Pass / Certification Acceptance Criteria
Biometric Match Performance	Edge Attendance Kiosk, MobileFaceNet Engine ¹⁰ .	Offline test harness with 10,000 face samples across diverse lighting conditions.	$FAR \leq$, $FRR \leq$, Anti-Spoofing Detection $\geq$.
Offline Synchronization	Mobile LMS Client, SQLite/Room Database, Sync Gateway ¹⁰ .	72-hour simulated WAN disconnect; generation of 500 local events, followed by network reconnection.	Zero record loss; zero data corruption; full state synchronization completed within 60 seconds ¹⁰ .

Credential Verification	Credential Signer, DigiLocker Pull/Push API ¹¹ .	Automated issuance, signature validation, schema parsing, DigiLocker pull handshake tests ¹² .	100% compliance with W3C VC 2.0 schema; public key verification success rate = 18.
AI Recommendation Reliability	Indic RAG Chatbot, Multi-Factor Job Matcher ¹⁰ .	1,000 synthetic career queries and resumes evaluated against ground-truth curriculum docs ¹ .	Zero hallucinated policy outputs; candidate matching accuracy score $r \geq$.
Load and Stress Testing	API Gateway, Relational Persistence Layer, Cache Cluster ¹⁰ .	Distributed load testing simulating 25,000 concurrent sessions submitting requests simultaneously ¹⁰ .	95th percentile latency $\leq$; zero unhandled HTTP 5xx errors; server CPU load $\leq$.

Implementation Roadmap and Strategic Phasing

Deploying an integrated ecosystem across NCCT's decentralized institutional landscape requires a structured, multi-phase implementation roadmap designed to maintain administrative continuity¹.

The first phase establishes the cloud-based enterprise ERP core across NCCT headquarters, VAMNICOM, the five RICMs, and fourteen ICMs¹. During this stage, training catalogs are digitized, cooperative societies transition to the online nomination portal, and dynamic QR code attendance is introduced via instructor mobile applications¹. Campus logistics modules are activated across all residential training facilities¹⁰.

The second phase expands edge capabilities, multilingual learning, and digital credentialing¹⁰. Edge attendance kiosks equipped with dual RGB+NIR cameras are installed across training centers¹⁰. The offline-first mobile LMS is launched for Android devices, delivering localized instructional modules on PACS Computerization and the Common Accounting System (CAS)⁶. The platform integrates with Bhashini ULCA translation pipelines and connects to DigiLocker and API Setu, enabling cryptographically verified digital certification¹⁰.

The final phase activates advanced intelligence services and expands ecosystem connectivity¹⁰. The Indic RAG career guidance chatbot is deployed across mobile and web interfaces, providing voice-enabled career counseling to rural youth¹⁰. The cooperative employment exchange is launched, onboarding PACS, cooperative banks, dairy unions, and

agricultural federations to post verified job openings⁵. The semantic candidate-job matching algorithm connects certified talent directly with cooperative employers, while data bridges link candidate profiles to the National Career Service (NCS) and Skill India Digital Hub (SIDH)¹⁰. In this phase, edge infrastructure also extends to the 109 Junior Cooperative Training Centres (JCTCs) across India⁴.

Project Phase	Implementation Window	Core Technical Deliverables	Participating Institutions	Target Milestone Gate
Phase 1: Enterprise ERP Core	Months 1 – 6	Central ERP deployment; society nomination portal; dynamic QR attendance; hostel logistics module ¹⁰ .	NCCT HQ, VAMNICOM Pune, 5 RICMs, 14 ICMs ¹ .	100% paperless registration; conflict-free scheduling engine operational ¹⁰ .
Phase 2: Edge Biometrics & LMS	Months 7 – 12	Edge facial kiosks; offline-first mobile LMS; Bhashini pipeline; DigiLocker credential issuance ¹⁰ .	20 Primary Training Campuses; Pilot PACS Batches ¹ .	Biometric verification operational; verifiable credentials active in DigiLocker ¹⁰ .
Phase 3: AI & Employment Exchange	Months 13 – 18	Indic career chatbot; cooperative job exchange; NCS/SIDH bridges; JCTC edge rollout ¹⁰ .	All 20 Institutes, ~109 JCTCs, Cooperative Employers ⁴ .	Multi-factor job matching active; live candidate placements in PACS and federations ⁶ .

Nuanced Operational Conclusions and Strategic Recommendations

The implementation of this integrated digital platform transitions cooperative capacity building

from a fragmented, paper-reliant workflow into an integrated, verified learning and employment ecosystem¹⁰. Unifying administrative ERP functions, offline-capable mobile instruction, and cryptographically verified certifications directly advances the Ministry of Cooperation's strategic vision of *Sahkar se Samriddhi* (Prosperity through Cooperation)¹. The project's architectural approach addresses the practical constraints of rural training delivery:

1. Using commercial off-the-shelf (COTS) edge devices for facial verification and QR scanning avoids dependencies on custom, proprietary hardware while maintaining operational continuity during network outages¹⁰.
2. The offline-first LMS architecture—built on local caching, monotonic CRDT state synchronization, and Bhashini multilingual speech pipelines—ensures consistent educational access for rural participants regardless of local connectivity limitations¹⁰.
3. The credentialing pipeline shifts verification from manual paper checks to cryptographically tamper-evident W3C Verifiable Credentials issued directly to citizen DigiLocker wallets, creating an authentic record of workforce competencies¹⁰.
4. Grounding the conversational career chatbot in verified institutional documentation through Retrieval-Augmented Generation (RAG) prevents speculative responses and provides rural learners with realistic career guidance⁶.
5. The semantic job-matching engine pairs technical competency scores with geographic distance decay factors, ensuring rural cooperatives can identify qualified local talent while providing trained youth with clear pathways to formal employment⁶.

To ensure lasting operational success, the platform must be supported by ongoing institutional commitments: maintaining edge hardware across rural training centers, updating course curricula to reflect evolving cooperative regulations, and enforcing data minimization and security controls in compliance with the Digital Personal Data Protection Act, 2023¹. By connecting institutional training with verifiable employment opportunities, the platform establishes an enduring digital foundation for the modernization of India's cooperative sector⁶.

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